

1. General Description

The EMS3905 is a dual SPDT low on-resistance analog switch. It can operate from 1.5V to 5.5V power supply. The device offers low ON-state resistance and excellent ON-state resistance matching with break-before-make feature, to prevent signal distortion during the transferring of a signal from one channel to another. The device is capable of truly isolation. Even when A overrides V_{CC} , very little current will flow back to the supply.

2. Features and Benefits

- Wide Supply Range from 1.5 V to 5.5 V
- Low ON-state resistance, typically 1.9Ω ($V_{CC}=5.5$ V)
- 1.8 V Compatible Control Logic Inputs
- High Off-Isolation: -80 dB @ 10 kHz
- Low Channel-to-Channel Crosstalk: -82 dB @ 10 kHz
- High Bandwidth (-3dB @ 800MHz) Suitable For USB2.0 High-Speed Routing
- Low Quiescent Current ($< 1\mu A$) when $V_{CC} = 3.3$ V

3. Applications

- Audio, Video, UART, USB2.0 Signal and Supply Routing
- Cell phones and TWS headset

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4. Ordering Information

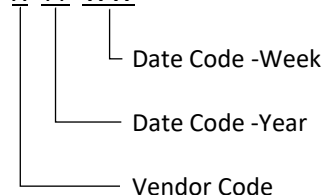
Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EMS3905MS	S3905 XYYWW	MSOP-10L	MSOP package, 10 pins 3.0mm × 3.0mm; 1.1 mm (Max) height	3000
EMS3905DRC	S3905 XYYWW	DFN3x3-10L	DFN package, 10 pins 3.0 mm × 3.0 mm; 0.8 mm (Max) height	5000
EMS3905RSW	3905 YYWW	QFN1.4x1.8-10L	QFN package, 10 pins 1.4 mm × 1.8 mm; 0.6 mm (Max) height	3000

MARKING INFORMATION

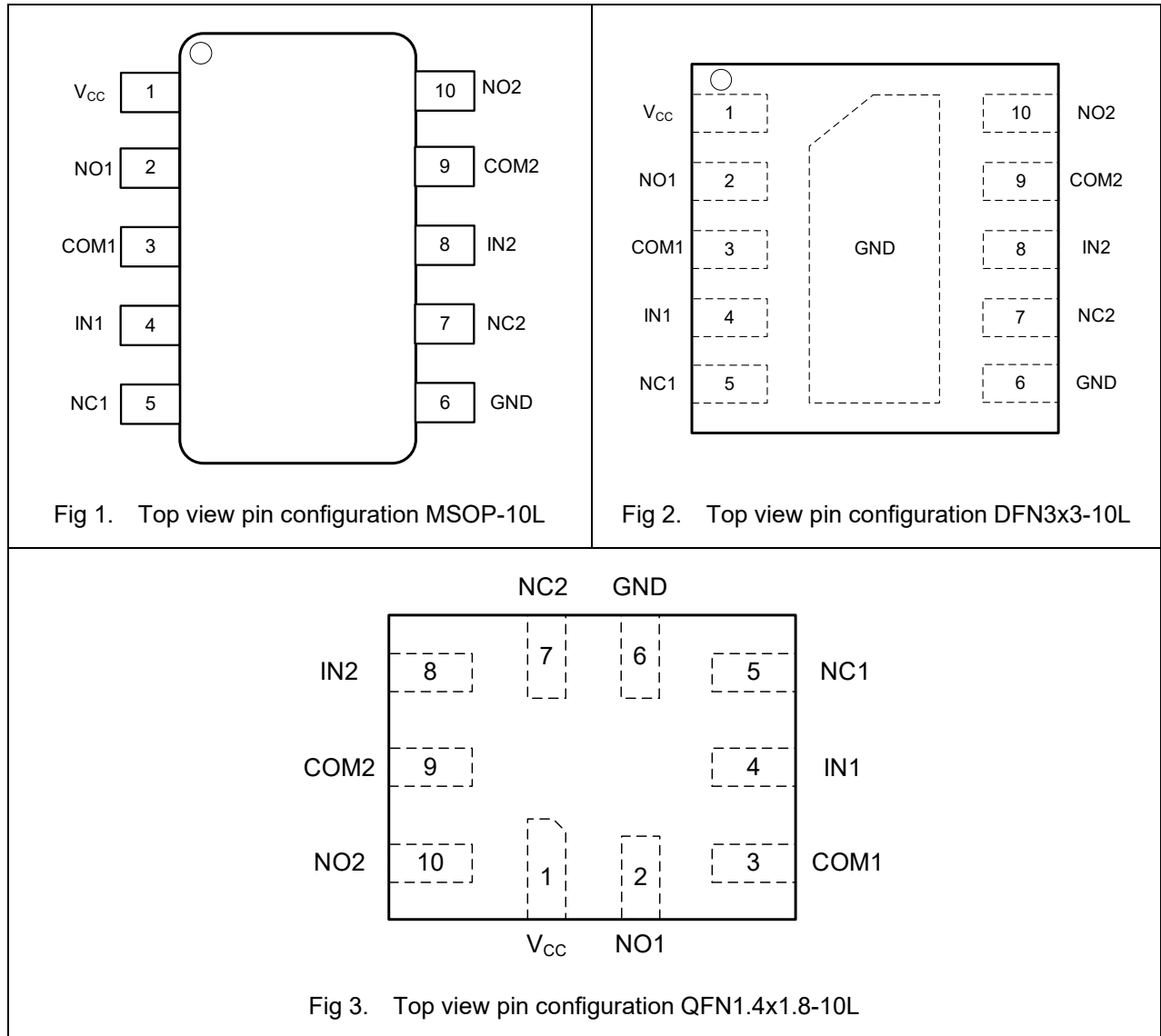
NOTE: XYYWW = Vendor Code and Date Code.

X YY WW



5. Pinning Information

5.1. Pin map



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5.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
	MSOP-10L/DFN3x3-10L/ QFN1.4x1.8-10L	
V _{CC}	1	Power supply pin
NO1	2	Normally open I/O for switch 1
COM1	3	Common I/O for switch 1
IN1	4	Select pin for switch 1
NC1	5	Normally closed I/O for switch 1
GND	6	Ground
NC2	7	Normally closed I/O for switch 2
IN2	8	Select pin for switch 2
COM2	9	Common I/O for switch 2
NO2	10	Normally open I/O for switch 2

6.Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level.

IN	NC TO COM, COM TO NC	NO TO COM, COM TO NO
L	ON	OFF
H	OFF	ON

7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V _{CC}	supply voltage		-0.3	6.5	V
V _{IN}	control Input Voltage		-0.3	6.5	V
V _A	signal input voltage		-0.3	6.5	V
I _A	continuous Current Through COM, NC, NO			±100	mA
	peak Current Through COM, NC, NO (pulsed at 1ms 50% duty cycle)			±200	mA
T _J	junction Temperature under Bias			150	°C
T _L	lead Temperature (Soldering, 10 seconds)			260	°C
R _{θJA}	thermal resistance			350	°C/W

8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Min	Typ	Max	Unit
V _{CC}	supply voltage	1.5	3.3	5.5	V
V _{IN}	control input voltage	0		5.5	V
V _A	signal input voltage	0		V _{CC}	V
T _{amb}	ambient temperature	-40	25	125	°C

9. Electrical Characteristics

Table 6. Static characteristics
 $T_{amb}=25\text{ }^{\circ}\text{C}$, $V_{CC}=3.3\text{ V}$, unless otherwise specified

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
DC Electrical Characteristics						
V_{IH}	input logic high level	$V_{CC} = 3.4\text{ V to }5.5\text{ V}$	1.6			V
		$V_{CC} = 1.5\text{ V to }3.3\text{ V}$	1.4			V
V_{IL}	input logic low level	$V_{CC} = 3.4\text{ V to }5.5\text{ V}$			0.6	V
		$V_{CC} = 1.5\text{ V to }3.3\text{ V}$			0.4	V
I_{CC}	supply current	$I_A = 0$, $V_{SEL} = 0$ or $V_{SEL} = V_{CC}$			1.0	μA
ΔI_{CC}	additional supply current	$I_A = 0$, $V_{CC} = 2.5\text{ V or }4.2\text{ V}$ $V_{SEL} = 1.6\text{ V}$			3.0	μA
I_{off}	OFF-state switch leakage current from COM to NC (or NO)	$V_A = 5\text{ V}$, V_{NC} or $V_{NO} = 0\text{ V}$			± 2.0	μA
R_{ON}	on-resistance	$V_{CC} = 3.5\text{ V}$, $V_A = 0.8\text{ V to }2.0\text{ V}$, $I_A = 10\text{ mA}$		3.4	4.0	Ω
		$V_{CC} = 4.5\text{ V}$, $V_A = 0.8\text{ V to }2.3\text{ V}$, $I_A = 10\text{ mA}$		2.4	3.0	Ω
		$V_{CC} = 5.5\text{ V}$, $V_A = 0.8\text{ V to }2.6\text{ V}$, $I_A = 10\text{ mA}$		1.9	2.5	Ω
R_{FLAT}	on-resistance flatness	$V_{CC} = 3.5\text{ V}$, $V_A = 0.8\text{ V to }2.0\text{ V}$, $I_A = 10\text{ mA}$		0.2	0.5	Ω
		$V_{CC} = 4.5\text{ V}$, $V_A = 0.8\text{ V to }2.3\text{ V}$, $I_A = 10\text{ mA}$		0.25	0.5	Ω
		$V_{CC} = 5.5\text{ V}$, $V_A = 0.8\text{ V to }2.6\text{ V}$, $I_A = 10\text{ mA}$		0.3	0.5	Ω
ΔR_{ON}	on-resistance matching between channels	$V_{CC} = 3.5\text{ V}$, $V_A = 0.8\text{ V to }2.0\text{ V}$, $I_A = 10\text{ mA}$		0.1	0.2	Ω
		$V_{CC} = 4.5\text{ V}$, $V_A = 0.8\text{ V to }2.3\text{ V}$, $I_A = 10\text{ mA}$		0.1	0.2	Ω
		$V_{CC} = 5.5\text{ V}$, $V_A = 0.8\text{ V to }2.6\text{ V}$, $I_A = 10\text{ mA}$		0.1	0.2	Ω

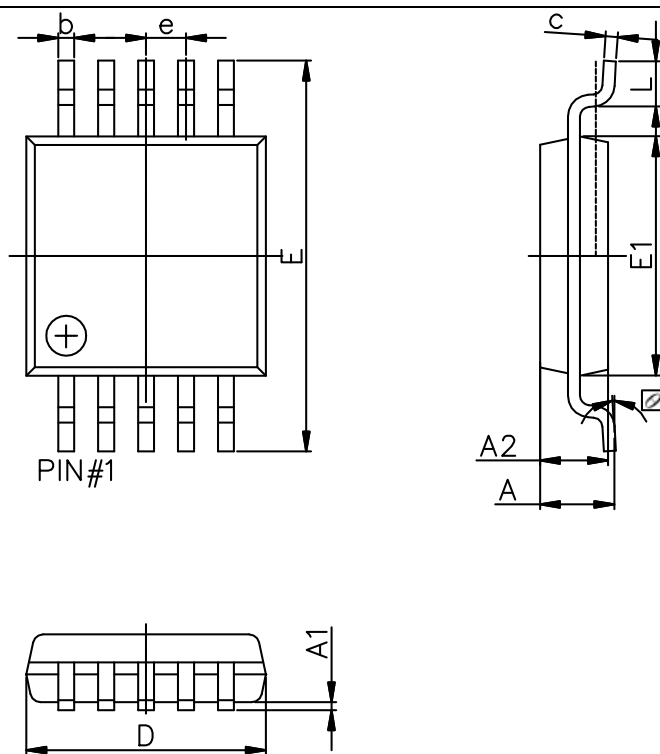
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Symbol	Parameter	Conditions	Min	Typ	Max	Unit
AC Electronics Characteristics						
t_{ON}	turn-On Time	$V_A=1.5V, C_L=50pF, R_L=50\Omega$		470		ns
t_{OFF}	turn-Off Time	$V_A=1.5V, C_L=50pF, R_L=50\Omega$		120		ns
t_{BBM}	break-before-make time	$V_A=1.5V, C_L=50pF, R_L=50\Omega$		600		ns
BW	-3dB Bandwidth	$C_L=0pF$		800		MHz
OIRR	off isolation	$f=10\text{ KHz}$		-80		dB
		$f=240\text{ MHz}$		-31		dB
X_{TALK}	crosstalk	$f=10\text{ KHz}$		-82		dB
		$f=240\text{ MHz}$		-26		dB
THD	total harmonic distortion	$f=20\text{Hz to } 20\text{KHz}$ $V_A=600mV_{pp}, R_L=32\Omega$		-80		dB
Capacitance						
C_{ON}	On Capacitance	$f=100\text{KHz}$		12		pF
C_{OFF}	Off Capacitance	$f=100\text{KHz}$		4		pF

10. Package Outline

MSOP-10L

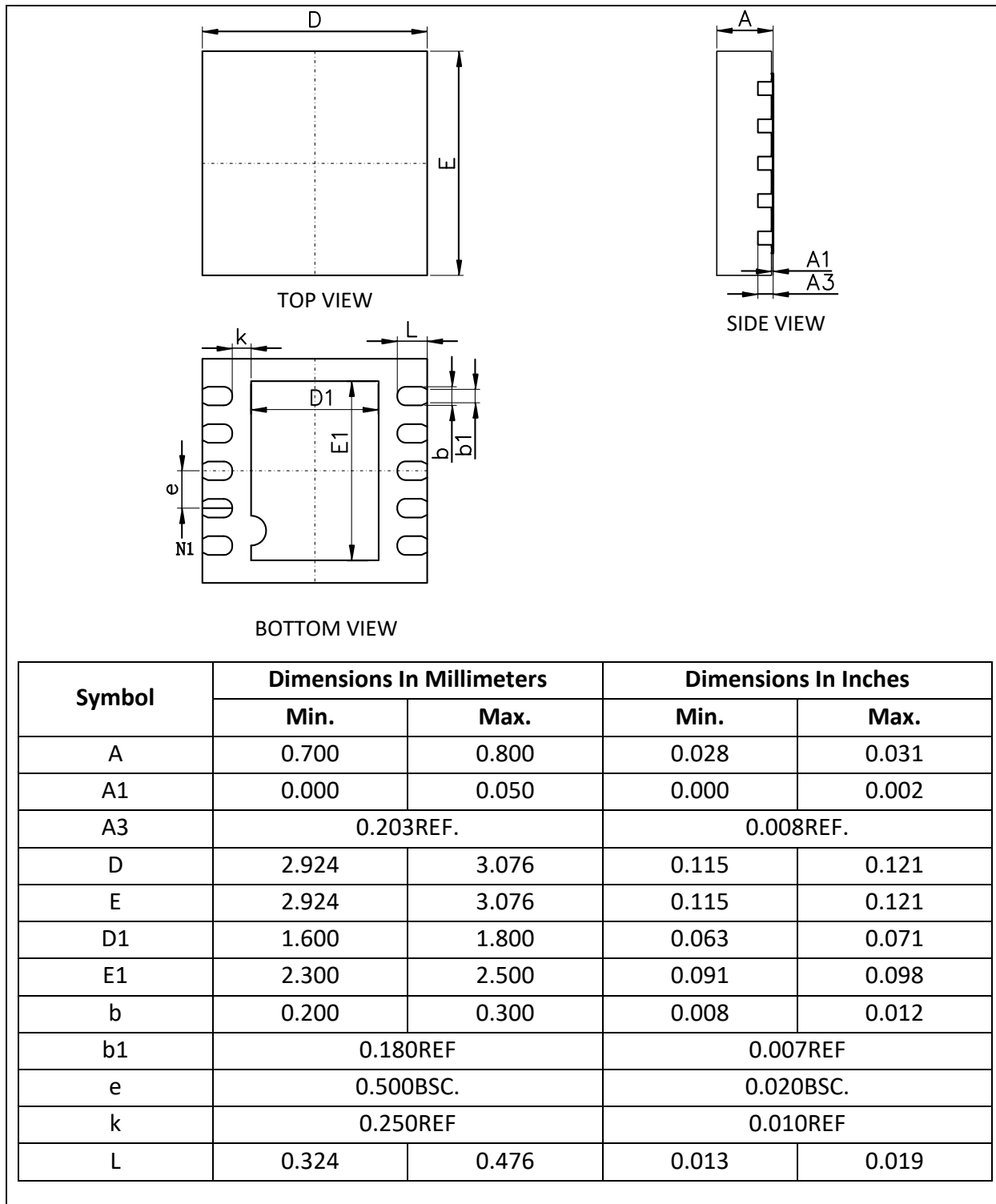


Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	---	1.100	---	0.043
A1	0.020	0.150	0.001	0.006
A2	0.750	0.950	0.030	0.037
b	0.180	0.330	0.007	0.013
c	0.090	0.230	0.004	0.009
D	2.900	3.100	0.114	0.122
e	0.500(BSC)		0.020(BSC)	
E	4.750	5.050	0.187	0.199
E1	2.900	3.100	0.114	0.122
L	0.400	0.800	0.016	0.031
θ	0°	6°	0°	6°

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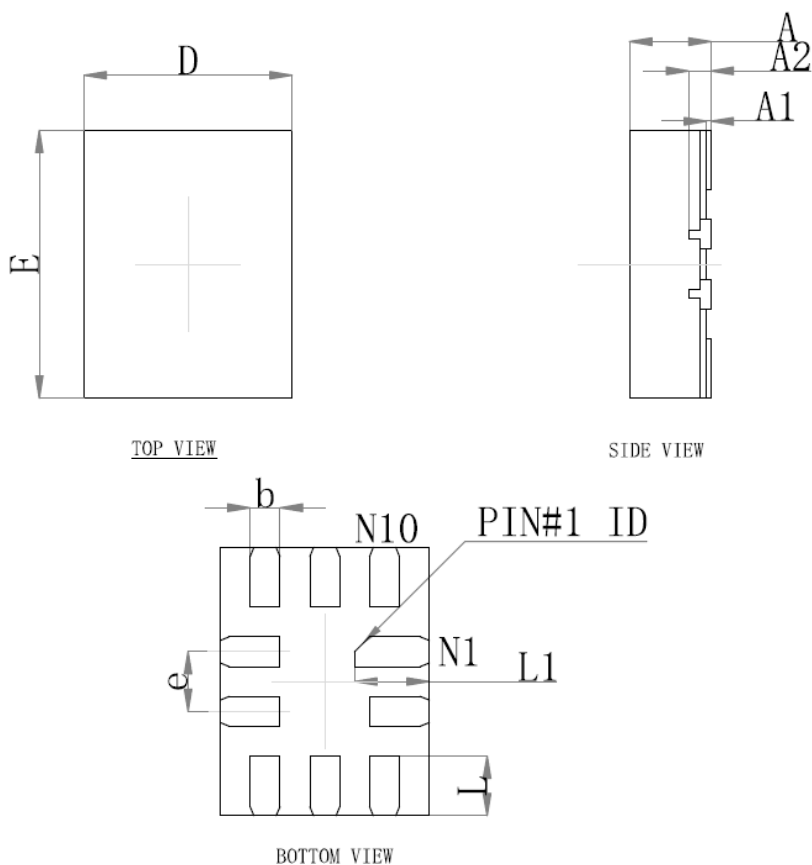
DFN 3x3 -10L



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QFN 1.4x1.8 -10L



Symbol	Dimension in Millimeters		
	Min.	NOM.	Max.
A	0.500	0.550	0.600
A1	0.000	--	0.050
A2	0.152 Ref.		
D	1.324	1.400	1.476
E	1.724	1.800	1.876
b	0.150	0.200	0.250
e	0.400 Typ.		
L	0.350	0.400	0.450
L1	0.450	0.500	0.550

11. Tape and Reel Information

11.1. Carrier tape dimensions

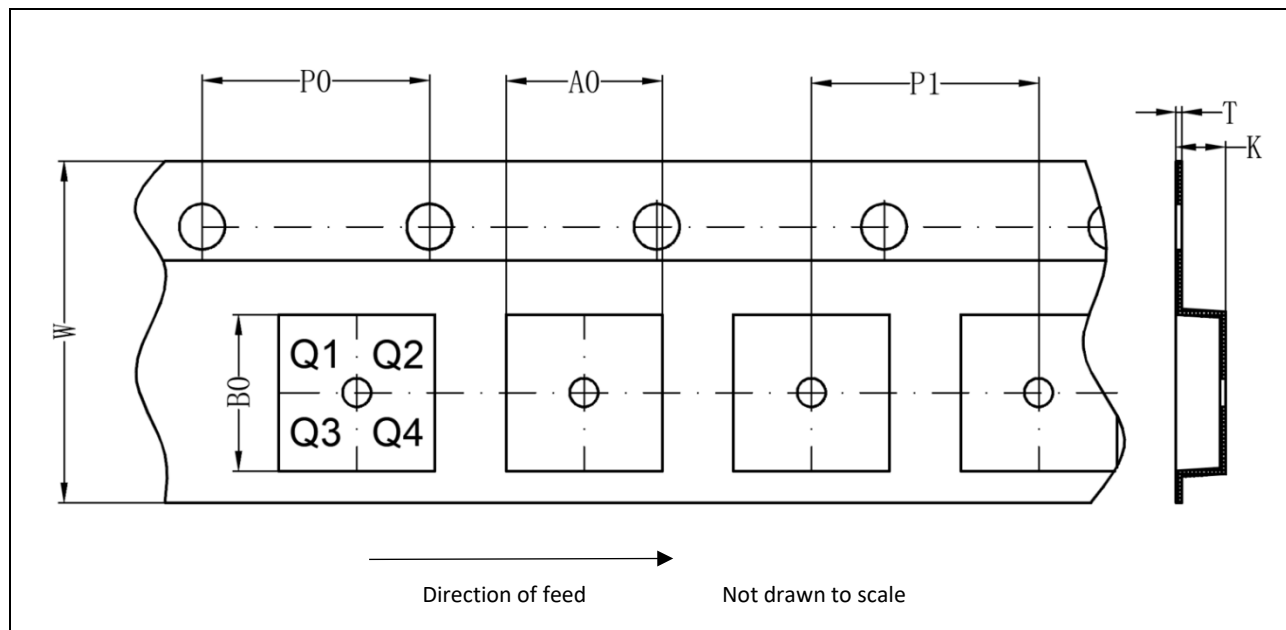


Table 10. Carrier tape dimensions

Package version	A0(mm)	B0(mm)	K(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
MSOP-10L	5.3	3.4	1.4	0.25	8	12	4	Q1
DFN3x3-10L	3.35	3.35	1.13	0.25	8	12	4	Q1
QFN1.4x1.8-10L	1.6	2.0	0.85	0.2	4	8	4	Q1

11.2. Reel and box dimensions

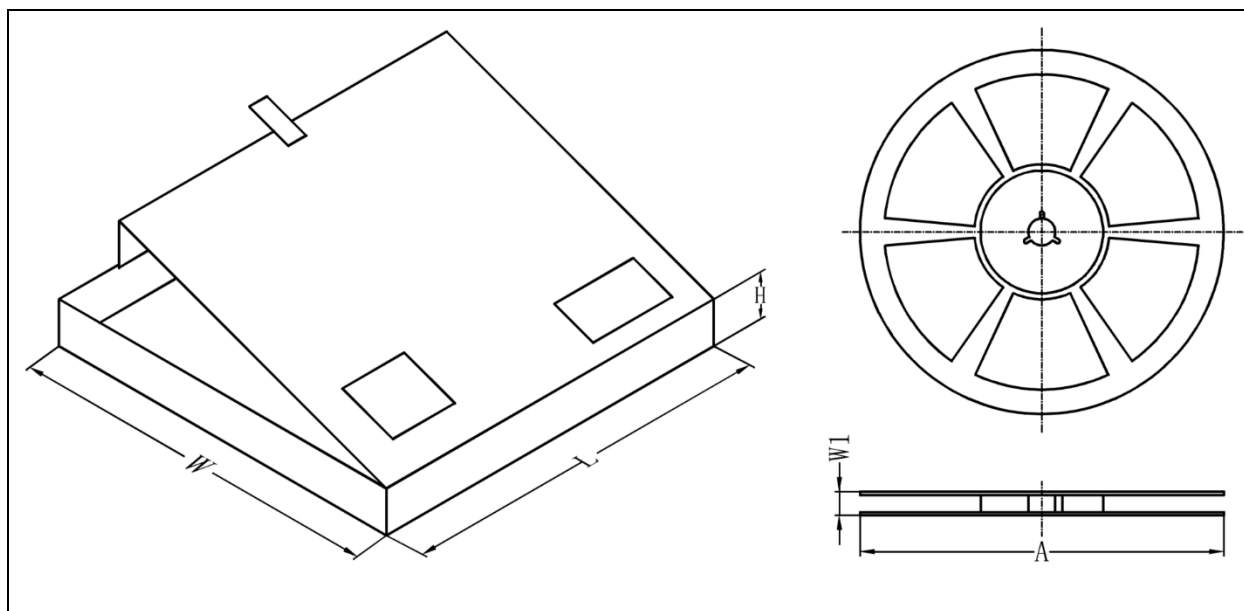


Table 11. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Max Reel Width W1 (mm)	SPQ (pcs)[1]	Reels per box	Outer box dimensions L×W×H(mm)[2]
MSOP-10L	MS	330	18.4	3000	1	358x340x50
DFN3x3-10L	DRC	330	18.4	5000	1	358x340x50
QFN1.4×1.8-10L	RSW	180	13.2	3000	1	210x200x40

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

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12. Revision History

Table 7. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EMS3905 Rev. 1.3	Apr 15, 2026	Product datasheet		EMS3905 Rev. 1.2
Modifications:	- Table 1 ordering information updated. - Section 11: Added Tape and Reel Information.			
EMS3905 Rev. 1.2	Jun 10, 2024	Product datasheet		EMS3905 Rev. 1.1
Modifications:	- Table2 updated. - Table6: R_{FLAT} updated.			
EMS3905 Rev. 1.1	Apr 10, 2024	Product datasheet		EMS3905 Rev. 1.0
Modifications:	Added the QFN package outline.			
EMS3905 Rev. 1.0	Feb 13, 2024	Product datasheet		